Journal of Chemical Neuroanatomy 120 (2022) 102069

Contents lists available at ScienceDirect

Journal of Chemical Neuroanatomy

journal homepage: www.elsevier.com/locate/jchemneu

Combination of *Ocimum sanctum extract* and Levetiracetam ameliorates cognitive dysfunction and hippocampal architecture in rat model of Alzheimer's disease

H.S. Nandini [a], K.L. Krishna [a,*], Chinnappa Apattira [b]

[a] Department of Pharmacology, JSS College of Pharmacy, JSS Academy of Higher Education & Research, Sri Shivarathreeshwara Nagara, Mysuru 570015, Karnataka, India

[b] Centre for Excellence in Molecular Biology and Regenerative Medicine (CEMR, DST-FIST Supported Center), Department of Biochemistry (DST-FIST Supported Department), JSS Medical College, JSS Academy of Higher Education & Research, Mysuru 570015, India

A R T I C L E I N F O

Keywords:
Ocimum sanctum
Epilepsy
Amyloid β
Neurodegenerative diseases
BDNF
Acetylcholinesterase
Cognitive dysfunction
Glutamate
GABA

A B S T R A C T

Alzheimer's disease (AD) is the most common neurodegenerative disease which affects more than 40 million people worldwide with progressive loss of memory and cognitive functions. It is reported, persistent AD is also one of the main causes of epilepsy in elders and comorbidity of both these will contribute to worsening the health status of AD patients. Recently, herbal plants with potent neuroprotective and antioxidant properties were used for increasing the quality of life in neurodegenerative disease patients. The present study was conceptualized to access the protective effect of *Ocimum sanctum* extract (OSE) and Levetiracetam (LEV) and their combination (OSE+LEV) against AD and epilepsy associated with AD in the rat AD model. AD was induced in aged male Wistar albino rats with Amyloid-β (Aβ) by intracerebroventricular administration. The results reveal, treatment with OSE, LEV and OSE+LEV significantly reversed the memory impairment, increases the BDNF expressions and decreases AChE activity in Aβ induced AD animals. Expression of A-β and p-tau in the hippocampus was significantly reduced in treatment group when compared to the control animals. Treatment with OSE and OSE+LEV also restored the hippocampal architecture by ameliorating the neuronal count in CA1, CA3 and DG regions. It also observed that treatment has decreased the excitoneurotoxicity evidenced by decreased glutamate and increased GABA levels and thus provided protection against epilepsy. Treatment groups also exhibited a potent antioxidant activity when tested endogenous antioxidant enzymes SOD, GSH and LPO in the brain hippocampus. Our findings provide evidence for use of OSE, LEV and OSE+LEV against AD and epilepsy associated with AD in Aβ induced AD animal model. However, further clinical studies are required to prove the use of OSE, LEV and OSE+LEV in the management of AD and AD-associated epilepsy in human volunteers.

1. Introduction

Alzheimer's disease (AD) and epilepsy are common disorders in the elder population and AD may be one of the main causes of developing epilepsy. Compared to other age groups, AD and epilepsy are seen more in the elderly population (Friedman et al., 2011; Beghi and Giussani, 2018; Liu et al., 2016). Due to the increased lifespan of human beings, the number of elderly populations is increasing globally and these age groups are predominantly affected by dementia, especially AD that can be attributed to 50–56% of cases at autopsy and clinical series. A large number of studies state that there is a paramount risk factor for developing epilepsy and seizures in dementia (Horváth et al., 2018) and the prevalence of epilepsy in AD is 10–20%(Larner, 2010; Jenssen and Schere, 2010; Hommet et al., 2008; Mendez and Lim, 2003). Co-morbidity of epilepsy associated with AD will contribute to the reduction in cognitive abilities and autonomy and increase the progression of AD (Miranda and Brucki, 2014).

Hippocampal altered network and impaired cellular functions are well reported in a different AD mouse model which contributes to spatial memory function and deficits in gamma-aminobutyric acid (GABA) pathways. These alterations of inhibition and excitation balance lead to seizures and help to understand epilepsy co-morbidity in AD patients.

* Correspondence to: Department of Pharmacology, JSS College of Pharmacy, JSS Academy of Higher Education & Research, Sri Shivarathreeshwara Nagara, Mysuru 570015, Karnataka, India.

E-mail addresses: hsnandini@jssuni.edu.in (H.S. Nandini), klkrishna@jssuni.edu.in (K.L. Krishna), auchinnapa16@gmail.com (C. Apattira).

https://doi.org/10.1016/j.jchemneu.2021.102069
Received 26 August 2021; Received in revised form 25 December 2021; Accepted 26 December 2021
Available online 29 December 2021
0891-0618/

This can be prevented by partially blocking NMDA receptor activity or by increasing in GABA and by decreasing the accumulation of amyloid-β plague, which is directly co-related to neuronal hyperactivity (Frontiers, 2021).

Since anti-epileptic drugs (AEDs) can stimulate or inhibit the hepatic metabolism of cytochrome p450 enzymes, they can cause a considerable amount of interaction as well as side effects with a large number of medications, and some of the AEDs can control the epileptic seizures but they contribute to cognitive dysfunction (Belcastro et al., 2007; Larner, 2010).

Ocimum sanctum (Tulsi) is known as a queen of herbs for its spiritual and restorative properties (Pattanayak et al., 2010). It contains volatile oils like eugenol, luteolin, apigenin and ursolic acid which are potentially being explored for drug development (Venuprasad et al., 2017 Jul). *Ocimum sanctum* has been reported to possess anti-inflammatory (Kadian and Parle, 2012), antioxidant effect (Ramesh and Satakopan, 2010 Jul), anti-allergic (Gupta et al., 2002 Jul) and reproductive medicine (Elmi et al., 2017). *Ocimum sanctum* extract (OSE) promotes the density of pyramidal cells in the CA1 and CA3 mediated by increased concentration of Ach (Kusindarta et al., 2018; Joshi and Parle, 2006). *Ocimum sanctum* combats oxidative stress and regulates neurotransmitter levels which are critical for neuronal function (Venuprasad et al., 2017 Jul).

Levetiracetam (LEV) is a newer drug used in the treatment of epilepsy and it has potential benefits in neurological and psychiatric conditions like AD as a nootropic and autism, bipolar disorder and anxiety disorders (Mohan et al., 2014). A high dose of LEV may lead to acute kidney injuries (Erdinc, et al.) and induce neurotoxicity. Thus, in the present study, we have evaluated excitotoxicity produced due to excessive accumulation of Aβ and p-tau in Aβ induced AD rat model. The protective effect of OSE and combination of OSE with LEV against cognitive dysfunction in seizures associated with AD by decreasing the accumulation of A-beta and p-tau and by also ameliorating cognitive dysfunction and hippocampal architecture.

2. Material and methods

2.1. Animals

Male Wistar Albino Rats were procured from Biogen Pvt. Ltd. Bengaluru and acclimatized for one week. The investigational procedure was approved by Institutional Animal Ethics Committee, JSS College of Pharmacy, Mysuru with approval number-JSSCPM/IAEC 360/2019.

2.2. Chemicals

Procured β-amyloid (1−42) from Bi Biotech India Pvt. Ltd, p-tau from cell signaling technology, brain-derived neurotrophic factor (BDNF) & Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) from a cloud clone corporation (USA), gamma-aminobutyric acid & glutamate from Sigma Aldrich chemical, India.

2.3. In-vivo studies

Standard procedure was used for bilateral Intracerebroventricular (ICV) administration of 5 μl of β-amyloid 1–42 (1 μg/ml) to both the right and left cerebral ventricles using coordinates: 0.8 mm posterior to bregma; 1.5 mm lateral to sagittal suture; 4 mm beneath from the surface of the skull (Cetin et al., 2013).

Male Rats were randomly divided into 6 groups each containing 8 animals. Group I: Normal (received vehicle for 42 days p.o), Group II: Sham control (received ICV infusion of ACSF + vehicle for 42 days p.o), Group III: Control (received ICV infusion of 5 μl β-amyloid + vehicle for 42 days p.o), Group IV: OSE (received ICV infusion of β- amyloid + OSE 400 mg/kg, p.o for 42 days) (Mirje et al., 2014) Group V: LEV (received ICV infusion of β- amyloid + LEV 400 mg/kg, p.o for 42 days) (Song et al., 2018) and Group VI: OSE + LEV (received ICV infusion of β-amyloid A-β + received combination of OSE + LEV (200 mg/kg, p.o) for 42 days.

2.4. Assessment of memory impairment

The Morris water maze (MWM) paradigm was used to access the spatial orientation of learning and memory, as described previously. The escape latency time (ELT) and time spends in the target quadrants (TSTQ) was recorded as an index of memory retrieval (Yang et al., 2014; Sachdeva et al., 2014).

2.5. Assessment of endogenous antioxidant and AChE

On 43rd day, animals were sacrificed; brain tissue was quickly collected and stored at −80°c. Brains hippocampus region was thawed and homogenized with ten times (w/v) ice-cold phosphate buffer solution (P^H 7.4) for biochemical estimation viz SOD, LPO (Ellman et al., 1961; Ellman, 1959; Ohkawa et al., 1979), GSH (Ellman, 1959) and AChE by Ellman's method (Ellman et al, 1961).

2.6. Expression analysis

Hippocampus was homogenized with ice-cold radio immune precipitation assay (RIPA) buffer for protein estimation, and the homogenate was centrifuged at 14,000 rpm for 5 mins. The commercially available BCA kit was used to measure the amount of protein present in the sample. Samples were normalized to 50 μg protein and separated by SDS-PAGE and transferred onto PVDF membranes. Antibodies against p-tau (1:1000) from cell signaling technology and BDNF (1:1000) and GAPDH (1:7000) purchased from Cloud Clone Corporation (USA) were used. The immune reactive bands were visualized using appropriate antibodies, with an enhanced chemiluminescence reagent (Advansta) using chemiluminescence documentation system UVITECH. Later, the visualized bands were quantified using Image J software, and expression was represented in the arbitrary units normalized to internal control.

2.7. Neurotransmitters

Sample preparation procedures: Hippocampus homogenate was centrifuged at 7800x g with 15 volumes of methanol/water for 15 minutes at 4 °C and supernatants were stored at −20°C till completion of GABA and glutamate analysis by reported methods (Yang et al., 2014).

2.8. Histopathology

The harvested brain was fixed using 10% formalin solution for 72 h and then washed with water for 30 minutes; dehydrated by using ascending grades of alcohol, cleared with xylene, and embedded in paraffin. And brain sections were stained with Nissl staining and examined under a microscope with a digital camera.

2.9. Statistical analysis

All values are expressed as Mean±SEM and carried out by one-way ANOVA followed by Bonferroni's multiple comparison test with the aid of Graph Pad Prism Software version 5.

3. Results

3.1. Nootropic activity of OSE and LEV

Aβ treated rats took a significantly ($p < 0.05$) longer time to reach the platform and a reduction in the time spent in the targeted quadrant when compared to the normal group and sham control which indicates a memory impairment. Treatment with OSE, LEV and OSE+LEV produced a significant($p < 0.05$) reduction in the time taken to reach the platform and an increase in the time spent in the targeted quadrant (Fig. 2).

3.2. Endogenous antioxidants

Aβ treated rats showed a significant ($p < 0.05$) reduction in the concentration of brain SOD, GSH and increased LPO levels when compared to normal and sham control. Even through Treatment with OSE, LEV revered the activity but only OSE + LEV group has shown a significant ($p < 0.05$) activity when compare to Aβ alone treated group (Table 1).

3.2.1. Effect of OSE and LEV on AChE levels

The control group of animals showed a significant *($p < 0.05$)* increase in AChE levels when compared to the normal and sham control group. Whereas all the treatment groups have shown a significant ($p < 0.05$) reduction in AChE levels when compared to control animals (Table 1).

3.3. Expressions of Aβ, p-tau and BDNF

Animals treated with Aβ alone have shown a significant ($p < 0.05$) increase in A-β, p-tau expression and decreased BDNF expressions when compared to normal and sham control groups. A significant ($p < 0.05$) reduction in A-β and p-tau was seen in treatment animals, which evidenced treatment may increase the A-β & p-tau clearance. The same was observed in BDNF expression i.e increased expression in the treatment group when compared to A-β treated rats (Fig. 3).

3.4. Effect of drugs treatments on Neurotransmitters

Aβ rats showed a significant increase ($p < 0.05$) in glutamate levels when compared to the normal and sham control group which leads to neurotoxicity. Treatment groups showed a significant ($p < 0.05$) reduction in glutamate levels when compare to Aβ treated rats (Fig. 4a). Aβ treated rats showed a significant reduction ($p < 0.05$) in GABA levels when compared to a normal and sham control group which may lead to seizures. Treatment groups (OSE and OSE+LEV) showed a significant increase ($p < 0.05$) in GABA when compared to Aβ treated rats (Fig. 4b).

3.5. Histopathology

Disruption in hippocampus volume and significant loss of neurons in CA1, CA3 and DG regions was observed in control animals. Treatment with OSE, LEV and their combination demonstrated a significant increase in the neuronal count and reversed the disruption (Fig. 5).

4. Discussion

Many AD animal models have shown cognitive deterioration following Aβ infusion into rodent's brains (Faucher et al., 2015). Spatial memory impairment is most likely the result of elevated A-β concentrations at localized areas, as evidenced by A-β administration by i.c.v (Schmid et al., 2017) as well as hippocampus routes. It is reported that upon administration, A-β gets deposited, which leads to synaptic disruption and other effects, which have been found to produce spatial memory impairment in the hippocampal area (Xuan et al., 2012). In the present investigation, we found OSE, LEV and the combination-treated group have shown a significant increase in spatial memory when compare to Aβ alone treated group.

GSH is a significant antioxidant that regulates the intracellular redox environment and detoxifies ROS (Hammond et al., 2001; Dringen, 2000). Its intracellular balance has been proven to be essential for brain cell health and function (Dringen, 2000). Several animal investigations have steadily shown that brain GSH depletion causes enhanced ROS-associated brain damage (Jain et al., 1991; Wüllner et al., 1996; García et al., 2000). Several *in-vitro* and *in-vivo* investigations have shown that GSH has a neuroprotective role against a wide range of oxidative insults (Dringen and Hirrlinger, 2003; Mytilineou et al., 2002) and found neurons are more sensitive to ROS at the low level of GSH (Wüllner et al., 1996). GSH has been demonstrated in studies to be involved in mitigating the harmful effects of ROS in brain cells, and which leads to enhanced apoptotic signaling and as a result, neuronal death (Schulz et al., 2000). GSH has been implicated in the development of several aging-related neurodegenerative illnesses, including AD (Schulz et al., 2000; Pocernich and Butterfield, 2012). GSH level reductions linked with AD were observed in both *in vitro* (Ghosh et al., 2012) and *in-vivo* models of AD (Resende et al., 2008).

It is evidenced that lipid peroxidation is a key process of neurodegeneration in the Alzheimer's brain. An array of antioxidants phytopharmaceuticals protects against lipid peroxidation in the brain cell membrane by decreasing lipid peroxidation. It is also reported that Aβ induced oxidation of protein will be responsible for neurodegeneration in the AD brain (Jhoo et al., 2004).

In the present study, we observed there was a decreased level of SOD, GSH and increased levels of lipid peroxidation in control group of animals, which evidences neurodegeneration. ICV administration of Aβ might have caused neurodegeneration and these are reversed by OSE, LEV as well as their combination treatment. By this, we can clearly say, OSE and LEV are having a neuroprotective effect and these can be used to prevent neurodegeneration in AD and epilepsy-like disease conditions.

Neurotransmitters are chemical messengers in the brain that amplify and modulate signals. Abnormal changes in neurotransmitter levels have been linked to a variety of neurological illnesses, including

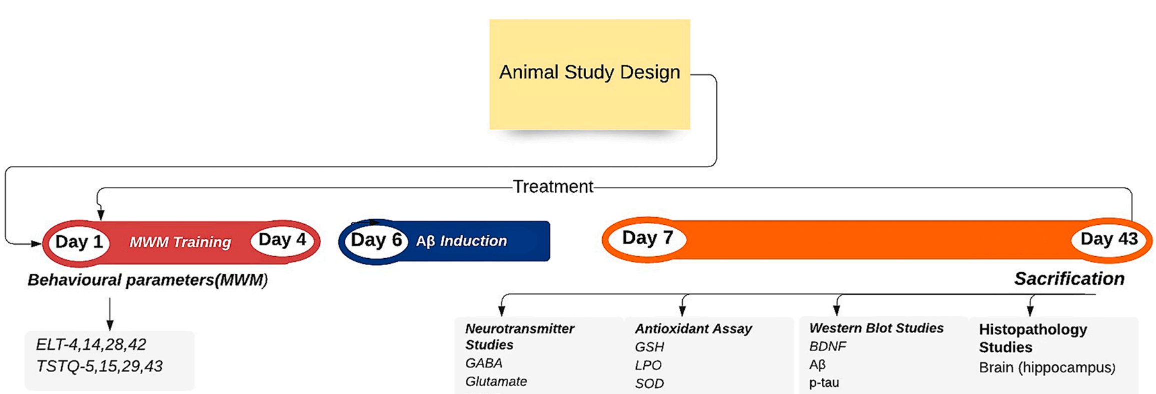

Fig. 1. Experimental study design *AChE: Acetylcholinesterase, SOD: Sodium dismutase, LPO: Lipid peroxidation, GSH Reduced glutathione, ELT: Escape latency time, TSTQ: time spends in the target quadrants, Aβ: Amyloid-beta, BDNF: brain-derived neurotrophic factor.*

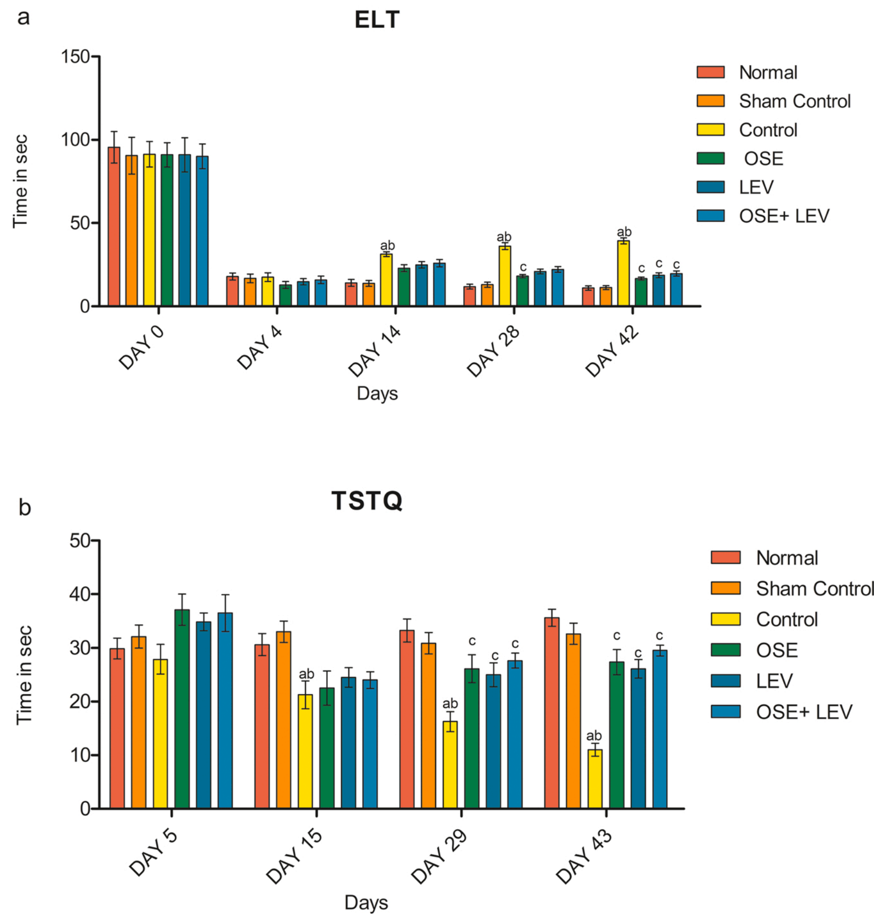

Fig. 2. Protective effect of OSE, LEV and in combination against A-β induced AD rat model (ELT and TSTQ) *N = 8, [a]P < 0.05, [b]P < 0.05, [c]P < 0.05, [d]P < 0.05- significantly different from the normal Vs control, sham control Vs control, treatment Vs control, OSE/LEV Vs OSE +LEV treated group respectively*.

Table 1
Protective effect of OSE, LEV and combination against A-β induced AD rat model (SOD, LPO, GSH and AChE levels).

Groups	SOD (μM/mg of protein)	LPO (nM/mg of protein)	GSH (μM/mg of protein)	AChE (μM/mg of protein/min)
Normal	11.03 ± 0.529	06.49 ± 0.652	12.85 ± 0.778	1.339 ± 0.055
Sham control	11.75 ± 0.573	06.35 ± 0.588	11.68 ± 1.270	1.461 ± 0.098
Control	05.29 ± 1.368[ab]	13.49 ± 1.045[ab]	06.74 ± 0.519[ab]	4.407 ± 0.683[ab]
OSE	09.65 ± 0.860	08.87 ± 1.592	10.68 ± 0.740	1.242 ± 0.171[c]
LEV	09.87 ± 0.879	09.81 ± 1.053	09.64 ± 1.073	1.375 ± 0.281[c]
OSE + LEV	10.65 ± 1.917[c]	07.00 ± 1.262[c]	11.88 ± 0.893[c]	1.298 ± 0.266[c]

SOD: Sodium dismutase, LPO: Lipid peroxidation, GSH: Reduced glutathione, AChE: Acetylcholinesterase.
N = 8, [a]P < 0.05, [b]P < 0.05, [c]P < 0.05, [d]P < 0.05- significantly different from the normal Vs control, sham control Vs control, treatment Vs control, OSE/LEV Vs OSE +LEV treated group respectively

epilepsy. Neurotransmitter regulation by epilepsy is one of the causes of epileptic seizures and cognition in clinical circumstances. Neurotransmitters play an important role in altering learning and memory function. GABA is a central nervous system inhibitory transmitter well recognized for its role in epilepsy (Shaikh et al., 2013). In the present study, the rats treated with Aβ had alone had lower levels of GABA, which contributes to epileptogenesis. GABA levels were found to be high in both the normal and treatment groups which demonstrated the important function of GABA in the management of epilepsy.

The etiology of epilepsy is closely linked to an increased excitatory amino acid, glutamate and its release set off a chain of events that ends in increased intracellular calcium and finally leads to neuronal death (Holmes, 2002). We observed increase level of glutamic acid in the control group of animals than to the normal group, which might contribute to neuronal death and reduced memory performance. The treatment with OSE, LEV and their combination reversed the elevated level of glutamate and thereby inhibited epileptogenesis.

Many biological activities associated with brain functions, such as learning and memory are altered by the manipulation of glutamate signaling. Acetylcholine, a signaling molecule elicits a variety of responses at CNS and neuromuscular junctions as well as playing an important role in controlling glutamate release and memory formation (Atzori et al., 2003). OSE also increases the expression of cholinergic and serotonergic neurotransmitters such as acetylcholine and serotonin, which play a pivotal role in memory consolidation. Inconsistency with the above study, we also found that OSE reduced the levels of AChE which might be responsible for improved memory function and which control excessive release of glutamate which may leads to excitotoxicity (seizures).

It is well reported that BDNF promotes the survival and development of various types of neurons. According to our findings, the level of BDNF expression in the control group was reduced whereas it was increased by

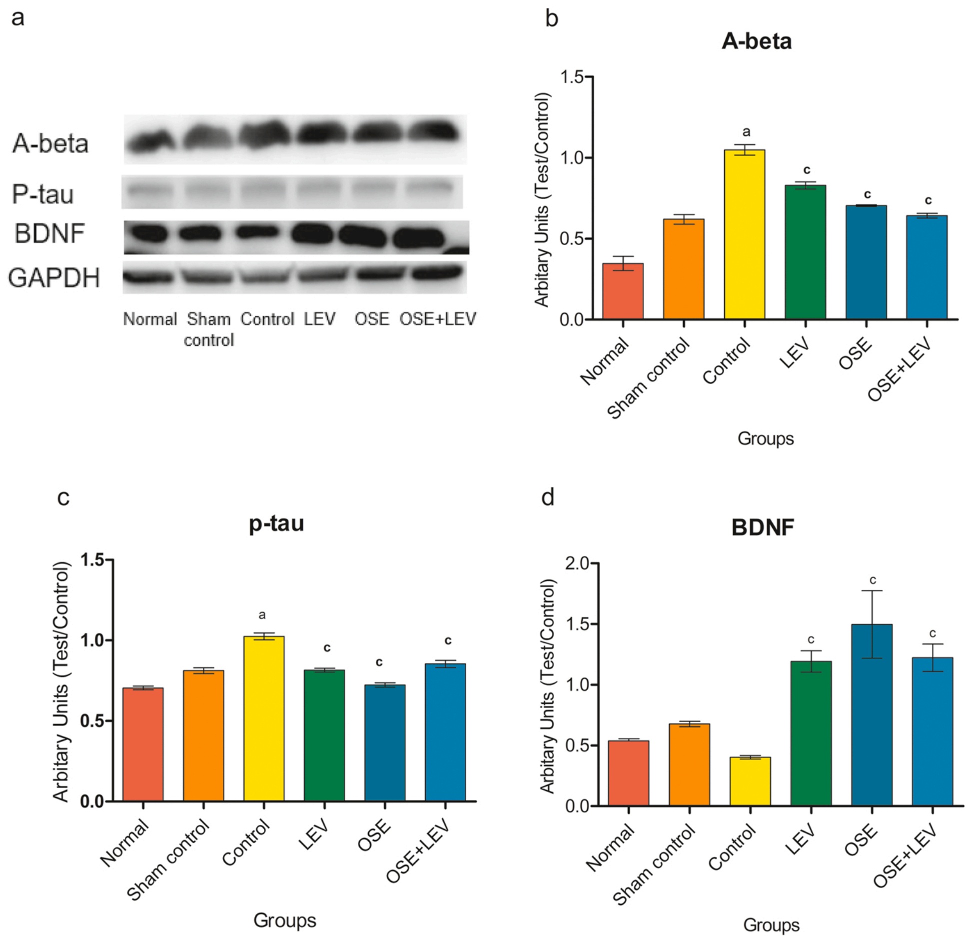

Fig. 3. Protective effect of OSE, LEV and OSE+LEV and in combination against A-β induced AD rat model (expression of A-β, p-tau and BDNF) Densitometric analysis of WB is shown as test/GAPDH ratio normalized to GAPDH *$n = 2$, $^{a}P < 0.05$, $^{b}P < 0.05$, $^{c}P < 0.05$, $^{d}P < 0.05$- significantly different from the normal Vs control, sham control Vs control, treatment Vs control, OSE/LEV Vs OSE +LEV treated group respectively*.

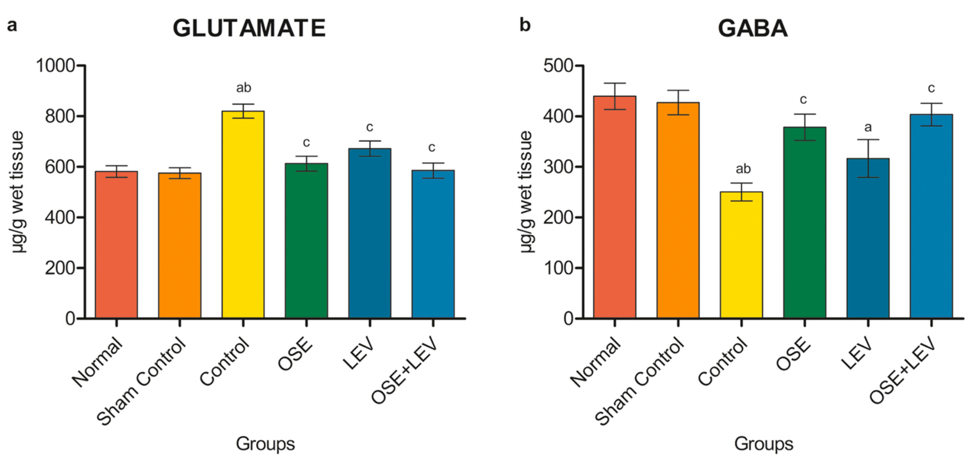

Fig. 4. Protective effect of OSE, LEV and its combination against A-β induced AD rat model (Glutamate and GABA levels). *$n = 8$, $^{a}P < 0.05$, $^{b}P < 0.05$, $^{cP} < 0.05$, $^{d}P < 0.05$- significantly different from the normal Vs control, sham control Vs control, treatment Vs control, OSE/LEV Vs OSE +LEV treated group respectively*.

OSE, LEV and their combination treatment.

p-tau and amyloid have been shown to spread across synaptically linked networks in the hippocampus(Cirrito et al., 2005; de Calignon et al., 2012). Neuronal hyperactivity in the hippocampus is co-related with Aβ levels and deficits in GABA pathways will alter the inhibition excitation balance leads to seizures. In the present study, we can see the accumulation of Aβ and p-tau in the control group when compare to normal which leads to malfunctioning of neuronal networks within the CA1, CA3, and dentate gyrus (DG) (Amaral et al., 2007) region.Which also leads to significant damage in the hippocampal architecture which leads to neuronal excitotoxcity and death. But in the treatment groups we can observed the reduction of Aβ accumulation and p-tau expressions and less neuronal damage in the hippocampus CA1, CA3 AND DG region then compare to control group. And it has also shown the protective

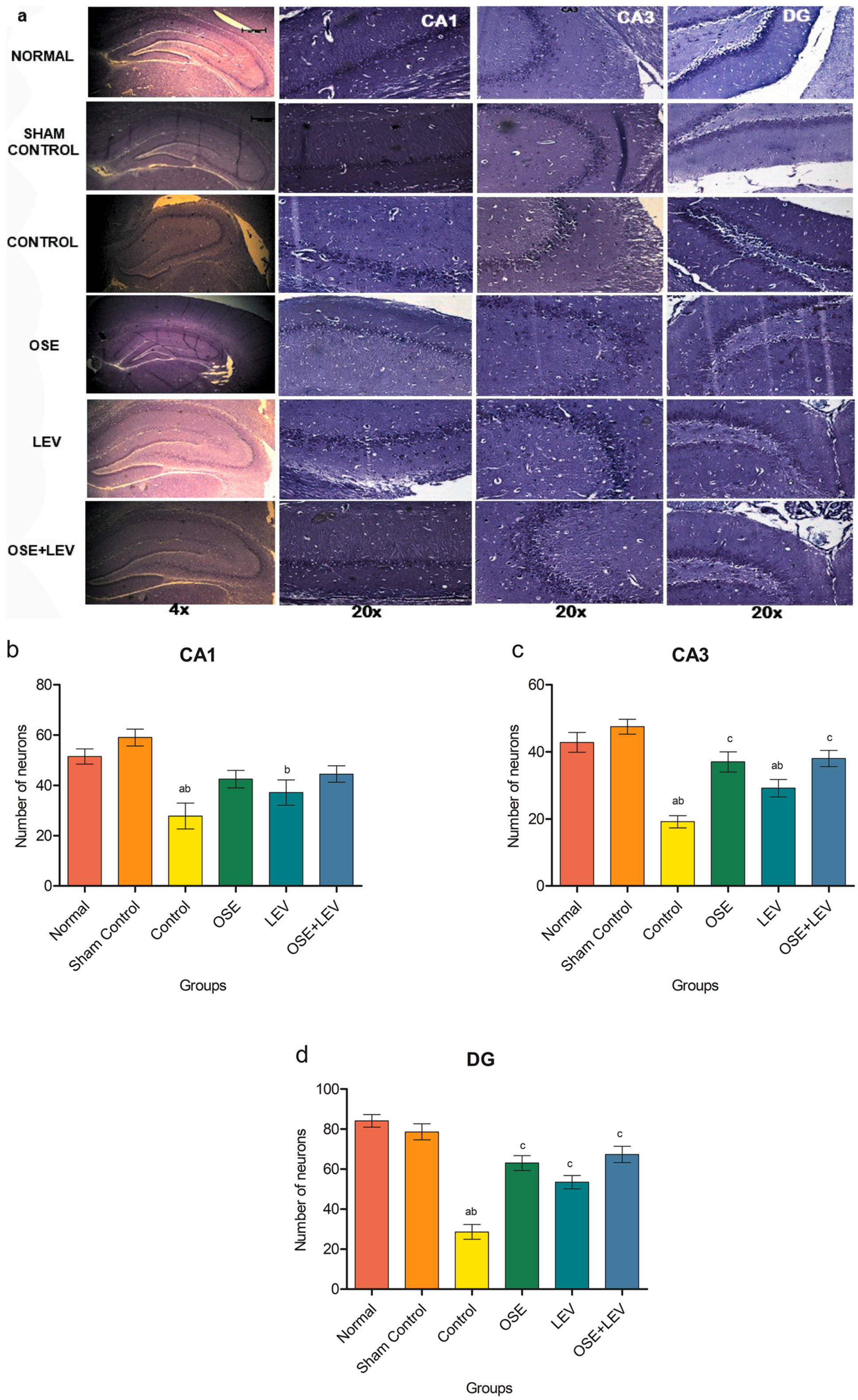

Fig. 5. Protective effect of OSE, LEV and in combination against A-β induced AD rat model (Photomicrograph of Nissl staining in CA1, CA3 and DG region of rat hippocampus) *$N = 3$, $^{a}P < 0.05$, $^{b}P < 0.05$, $^{c}P < 0.05$, $^{d}P < 0.05$- significantly different from the normal Vs control, sham control Vs control, treatment Vs control, OSE/LEV Vs OSE +LEV treated group respectively*.

effects against excitotoxcity(seizures) in AD rat model by decreasing the accumulation of Aβ and p-tau which were the main cause for damaged hippocampal architecture and excitotoxcity.

5. Conclusion

We conclude that ICV administration of Aβ in Wistar albino rats has produced AD and epilepsy-associated AD as demonstrated with behavioral, neurochemical and histopathological analysis. OSE, LEV and their combination showed protective effects against AD and epilepsy-associated AD by restoring the neurochemical and histological changes induced by Aβ. It was observed OSE and LEV can be good remedies against epilepsy associated with AD. Further detailed pre-clinical and clinical investigation may come up with novel remedies to treat AD and epilepsy-associated AD based on our findings.

CRediT authorship contribution statement

Nandini.H.S: Contributed to the design, data collection, interpretation, statistical analysis, and writing of the manuscript. **K.L.Krishna:** Contributed to designing and the critical evaluation of manuscript preparation., **Chinnappa Apattira:** Contributed in western blot studies.

Declaration of Competing Interest

All the authors declare no Conflict of interest.

Acknowledgment

The authors are thankful to the Principal, JSS College of Pharmacy, JSS Academy of Higher Education & Research, Mysuru 570015, Karnataka (India) for unconditional support and providing technical facilities for this work.

References

Amaral, D.G., Scharfman, H.E., Lavenex, P., 2007. The dentate gyrus: fundamental neuroanatomical organization (dentate gyrus for dummies). Prog. Brain Res. 163, 3–22.
Atzori, M., Kanold, P., Pineda, J.C., Flores-Hernandez, J., 2003. Dopamine-acetylcholine interactions in the modulation of glutamate release. Ann. N Y Acad. Sci. 1003, 346–348.
Beghi, E., Giussani, G., 2018. Aging and the Epidemiology of Epilepsy. NED 51 (3–4), 216–223.
Belcastro, V., Costa, C., Galletti, F., Pisani, F., Calabresi, P., Parnetti, L., 2007. Levetiracetam monotherapy in Alzheimer patients with late-onset seizures: a prospective observational study. Eur. J. Neurol. 14 (10), 1176–1178 (Oct).
de Calignon, A., Polydoro, M., Suárez-Calvet, M., William, C., Adamowicz, D.H., Kopeikina, K.J., et al., 2012. Propagation of tau pathology in a model of early Alzheimer's disease. Neuron. 73 (4), 685–697.
Cetin, F., Yazihan, N., Dincer, S., Akbulut, G., 2013. The effect of intracerebroventricular injection of beta amyloid peptide (1-42) on caspase-3 activity, lipid peroxidation, nitric oxide and NOS expression in young adult and aged rat brain. Turk Neurosurg. 23 (2), 144–150.
Cirrito, J.R., Yamada, K.A., Finn, M.B., Sloviter, R.S., Bales, K.R., May, P.C., et al., 2005. Synaptic activity regulates interstitial fluid amyloid-beta levels in vivo. Neuron. 48 (6), 913–922.
Dringen, R., 2000. Metabolism and functions of glutathione in brain. Prog. Neurobiol. 62 (6), 649–671.
Dringen, R., Hirrlinger, J., 2003. Glutathione pathways in the brain. Biol. Chem. 384 (4), 505–516.
Ellman, G.L., 1959. Tissue sulfhydryl groups. Arch. Biochem. Biophys. 82 (1), 70–77.
Ellman, G.L., Courtney, K.D., Andres, V., Featherstone, R.M., 1961. A new and rapid colorimetric determination of acetylcholinesterase activity. Biochem. Pharmacol. 7 (2), 88–95.
Elmi A., Ventrella D., Barone F., Filippini G., Benvenuti S., Pisi A., et al. Thymbra capitata (L.) Cav. and Rosmarinus officinalis (L.) Essential Oils: In Vitro Effects and Toxicity on Swine Spermatozoa. Molecules [Internet]. 2017 Dec 6 [cited 2020 Nov 23];22(12). Available from: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6149686/.
Erdinc B., Ghanta S., Andreev A., Elkholy KO, Sahni S. Acute Kidney Injury Caused by Levetiracetam in a Patient With Status Epilepticus. Cureus. 12(6):e8814.
Faucher, P., Mons, N., Micheau, J., Louis, C., Beracochea, D.J., 2015. Hippocampal injections of oligomeric amyloid β-peptide (1-42) induce selective working memory deficits and long-lasting alterations of erk signaling pathway. Front. Aging Neurosci. 7, 245.
Friedman, D., Honig, L.S., Scarmeas, N., 2011. Seizures and epilepsy in alzheimer's disease. CNS Neurosci. Ther. 18 (4), 285–294.
Frontiers | Hippocampal Deficits in Amyloid-β-Related Rodent Models of Alzheimer's Disease | Neuroscience [Internet]. [cited 2021 Aug 18]. Available from: https://www.frontiersin.org/articles/10.3389/fnins.2020.00266/full.
García, J.C., Remires, D., Leiva, A., González, R., 2000. Depletion of brain glutathione potentiates the effect of 6-hydroxydopamine in a rat model of Parkinson's disease. J. Mol. Neurosci. 14 (3), 147–153.
Ghosh, D., LeVault, K.R., Barnett, A.J., Brewer, G.J., 2012. A reversible early oxidized redox state that precedes macromolecular ROS damage in aging nontransgenic and 3xTg-AD mouse neurons. J. Neurosci. 32 (17), 5821–5832.
Gupta, S.K., Prakash, J., Srivastava, S., 2002. Validation of traditional claim of Tulsi, Ocimum sanctum Linn. as a medicinal plant. Indian J. Exp. Biol. 40 (7), 765–773.
Hammond, C.L., Lee, T.K., Ballatori, N., 2001. Novel roles for glutathione in gene expression, cell death, and membrane transport of organic solutes. J. Hepatol. 34 (6), 946–954.
Holmes, G.L., 2002. Seizure-induced neuronal injury: animal data. Neurology 59 (9 Suppl 5), S3–S6.
Hommet, C., Mondon, K., Camus, V., De Toffol, B., Constans, T., 2008. Epilepsy and dementia in the elderly. Dement. Geriatr Cogn. Disord. 25 (4), 293–300.
Horváth, A., Szűcs, A., Hidasi, Z., Csukly, G., Barcs, G., Kamondi, A., 2018. Prevalence, semiology, and risk factors of epilepsy in alzheimer's disease: an ambulatory EEG study. J. Alzheimers Dis. 63 (3), 1045–1054.
Jain, A., Mårtensson, J., Stole, E., Auld, P.A., Meister, A., 1991. Glutathione deficiency leads to mitochondrial damage in brain. Proc. Natl. Acad. Sci. U S A 88 (5), 1913–1917.
Jenssen, S., Schere, D., 2010. Treatment and management of epilepsy in the elderly demented patient. Am. J. Alzheimers Dis. Other Demen. 25 (1), 18–26.
Jhoo, J.H., Kim, H.-C., Nabeshima, T., Yamada, K., Shin, E.-J., Jhoo, W.-K., et al., 2004. Beta-amyloid (1-42)-induced learning and memory deficits in mice: involvement of oxidative burdens in the hippocampus and cerebral cortex. Behav. Brain Res. 155 (2), 185–196.
Joshi, H., Parle, M., 2006. Evaluation of nootropic potential of Ocimum sanctum Linn. in mice. Indian J. Exp. Biol. 44 (2), 133–136.
Kadian, R., Parle, M., 2012. Therapeutic potential and phytopharmacology of tulsi. Life Sci. 3 (7), 10.
Kusindarta, D.L., Wihadmadyatami, H., Haryanto, A., 2018. The analysis of hippocampus neuronal density (CA1 and CA3) after Ocimum sanctum ethanolic extract treatment on the young adulthood and middle age rat model. Vet. World 11 (2), 135–140.
Larner, A.J., 2010. Epileptic seizures in AD patients. Neuromolecular. Med. 12 (1), 71–77.
Liu, S., Yu, W., Lü, Y., 2016. The causes of new-onset epilepsy and seizures in the elderly. Neuropsychiatr. Dis. Treat 12, 1425–1434.
Mendez, M., Lim, G., 2003. Seizures in elderly patients with dementia: epidemiology and management. Drugs Aging 20 (11), 791–803.
Miranda, D. da C., Brucki, S.M.D., 2014. Epilepsy in patients with Alzheimer's disease: A systematic review. Dement Neuropsychol. 8 (1), 66–71.
Mirje MM, Zaman SU, Ramabhimaiah S. Evaluation of the anti-inflammatory activity of Ocimum sanctum Linn (Tulsi) in albino rats. 2014;8.
Mohan AJ, Kl K., Jisham KM, Mehdi S., Nidavani RB. Protective effect of tulsi and levetiracetam on memory impairment induced by pregabalin on mice. 2014;
Mytilineou, C., Kramer, B.C., Yabut, J.A., 2002. Glutathione depletion and oxidative stress. Parkinsonism Relat. Disord. 8 (6), 385–387.
Ohkawa, H., Ohishi, N., Yagi, K., 1979. Assay for lipid peroxides in animal tissues by thiobarbituric acid reaction. Anal. Biochem. 95 (2), 351–358.
Pattanayak, P., Behera, P., Das, D., Panda, S.K., 2010. Ocimum sanctum Linn. A reservoir plant for therapeutic applications: an overview. Pharmacogn. Rev. 4 (7), 95–105.
Pocernich, C.B., Butterfield, D.A., 2012. Elevation of glutathione as a therapeutic strategy in Alzheimer disease. Biochim Biophys Acta 1822 (5), 625–630.
Ramesh, B., Satakopan, V.N., 2010. Antioxidant activities of hydroalcoholic extract of ocimum sanctum against cadmium induced toxicity in rats. Indian J. Clin. Biochem. 25 (3), 307–310.
Resende, R., Moreira, P.I., Proença, T., Deshpande, A., Busciglio, J., Pereira, C., et al., 2008. Brain oxidative stress in a triple-transgenic mouse model of Alzheimer disease. Free Radic. Biol. Med. 44 (12), 2051–2057.
Sachdeva, A.K., Kuhad, A., Chopra, K., 2014. Naringin ameliorates memory deficits in experimental paradigm of Alzheimer's disease by attenuating mitochondrial dysfunction. Pharmacol. Biochem. Behav. 127, 101–110.
Schmid, S., Jungwirth, B., Gehlert, V., Blobner, M., Schneider, G., Kratzer, S., et al., 2017. Intracerebroventricular injection of beta-amyloid in mice is associated with long-term cognitive impairment in the modified hole-board test. Behav. Brain Res. 324, 15–20.
Schulz, J.B., Lindenau, J., Seyfried, J., Dichgans, J., 2000. Glutathione, oxidative stress and neurodegeneration. Eur. J. Biochem. 267 (16), 4904–4911 (Aug).
Shaikh, M.F., Sancheti, J., Sathaye, S., 2013. Effect of eclipta alba on acute seizure models: a GABAA-mediated effect. Indian J. Pharm. Sci. 75 (3), 380–384.
Song, H., Tufa, U., Chow, J., Sivanenthiran, N., Cheng, C., Lim, S., et al., 2018. Effects of antiepileptic drugs on spontaneous recurrent seizures in a novel model of extended hippocampal kindling in mice. Front. Pharmacol. 9, 451.
Venuprasad, M.P., Kandikattu, H.K., Razack, S., Amruta, N., Khanum, F., 2017. Chemical composition of Ocimum sanctum by LC-ESI-MS/MS analysis and its protective effects against smoke induced lung and neuronal tissue damage in rats. Biomed. Pharmacother 91, 1–12.
Wüllner, U., Löschmann, P.A., Schulz, J.B., Schmid, A., Dringen, R., Eblen, F., et al., 1996. Glutathione depletion potentiates MPTP and MPP+ toxicity in nigral dopaminergic neurones. Neuroreport. 7 (4), 921–923.
Xuan, A., Long, D., Li, J., Ji, W., Zhang, M., Hong, L., et al., 2012. Hydrogen sulfide attenuates spatial memory impairment and hippocampal neuroinflammation in β-amyloid rat model of Alzheimer's disease. J. Neuroinflamm. 9, 202.
Yang, W., Ma, J., Liu, Z., Lu, Y., Hu, B., Yu, H., 2014. Effect of naringenin on brain insulin signaling and cognitive functions in ICV-STZ induced dementia model of rats. Neurol Sci 35 (5), 741–751.